

- 4 (a) (i) noise/distortion is removed (from the signal) B1
the (original) signal is reformed/reproduced/recovered/restored B1 [2]
- or*
- signal detected above/below a threshold creates new signal (B1)
of 1s and 0s (B1)
- (ii) noise is superposed on the (displacement of the) signal/cannot be distinguished
or
analogue/signal is continuous (so cannot be regenerated)
or
analogue/signal is not discrete (so cannot be regenerated) B1
- noise is amplified with the signal B1 [2]

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(b) (i) $\text{gain/dB} = 10 \lg (P_2 / P_1)$

$$32 = 10 \lg [P_{\text{MIN}} / (0.38 \times 10^{-6})]$$

or

$$-32 = 10 \lg (0.38 \times 10^{-6} / P_{\text{MIN}})$$

C1

$$P_{\text{MIN}} = 6.0 \times 10^{-4} \text{ W}$$

A1 [2]

(ii) $\text{attenuation} = 10 \lg [(9.5 \times 10^{-3}) / (6.02 \times 10^{-4})]$

C1

$$= 12 \text{ dB}$$

$$\text{attenuation per unit length} (= 12/58) = 0.21 \text{ dB km}^{-1}$$

A1 [2]